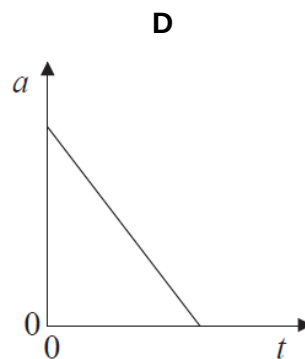
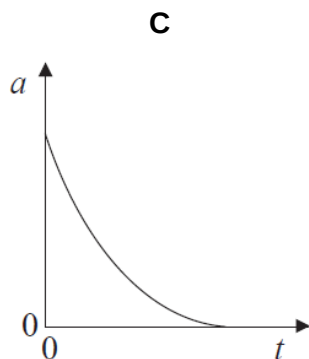
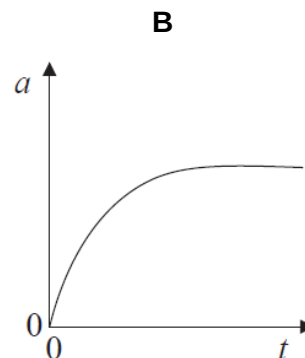
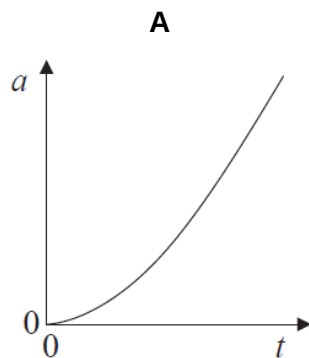


- 3 A ball, initially at rest, is dropped in the air from a great height. Air resistance is not negligible.

Which of the following graphs best shows the variation with time t of the acceleration a of the ball?



- | | | |
|----------|----------|--|
| 3 | C | $mg - kv = ma$
As time increases, a drops.
Since v is not linear with time because acceleration is not constant
therefore must be C |
|----------|----------|--|

[Turn over

4. A lift consists of a passenger car supported by a cable which runs over a light-frictionless